

EMPLOYMENT

Software Engineer, Performance

Automation

NetApp, Inc

2017 - 2026

- Accelerated environment staging times and eliminated manual setup errors by developing an end-to-end Python automation framework to configure complex NetApp storage systems and client (Linux / Windows) environments.
- Optimized performance automation bottlenecks up to 70% using distributed concurrent configuration.
- Maximized hardware utilization and streamlined team workflows using a Django web application that efficiently queues and schedules performance runs against constrained, shared laboratory resources.
- Expanded test coverage across diverse test requirements by integrating and deploying specialized workload generators designed to stress-test specific storage performance boundaries.

Software Engineer, Web Application

NetApp, Inc

2015 - 2017

- Unified disparate support metrics by designing and maintaining a custom PostgreSQL database tailored to ingest and organize complex NetApp customer support data.
- Uncovered new up-sell opportunities by developing data-mining algorithms within a Django web application to flag accounts primed for hardware or software upgrades.
- Boosted technical sales engagement by engineering an intuitive, social-media-inspired UI dashboard that surfaced critical technical bulletins and high-probability sales leads for account teams.

Software Engineer, Data Protection

NetApp, Inc

2013 - 2016

- Delivered a low-RPO data recovery solution by co-designing and implementing an IO synchronous replication system, enabling near-zero data loss capabilities for critical enterprise workloads.
- Ensured strict data integrity across distributed environments by programming kernel-level C++ software to replicate granular SCSI protocol commands between remote storage systems.
- Improved replication throughput and reduced latency overhead by engineering performance optimizations within the synchronous replication pipeline.

Component Lead, Scalable SAN

NetApp, Inc

2007 - 2013

- Guaranteed high availability and reliability for a core SAN configuration component by spearheading the software architecture, kernel-level C programming, and quality control guardrails for a clustered ONTAP management solution.
- Accelerated feature delivery by providing technical leadership and architectural mentorship to a software engineering team building multi-threaded, asynchronous software designs.
- Eliminated cross-team dependency bottlenecks and optimized project scheduling by managing workloads and spearheading cross-functional collaboration to resolve complex, cluster-wide engineering roadblocks.

Software Engineer

Adaptec

2003 - 2007

- Ensured seamless industry interoperability across Adaptec product lines by serving as the Technical Lead for the implementation of the Storage Management Initiative Specification (SMI-S)

Software Engineer

IBM

1999 - 2003

- Contributed to enterprise-level product stability and feature delivery by writing, debugging, and maintaining high-performance code within a collaborative software engineering team (Java, C++).

EDUCATION

Blacksburg, VA

Virginia Polytechnic Institute and State University

1994 - 1998

- B.S.E. Computer Engineering
- B.S.E. Minor: Computer Science, Mathematics

CORE SKILLS

- AI-Assisted development, prompt engineering for code development and optimization
- Clustered, multi-threaded, asynchronous, scalable, available, reliable, and supportable software
- Software architecture and technical team leadership
- Python, C++, C, Java, Perl, Databases
- Django web applications
- Storage technologies: SAN, iSCSI, Fibre Channel, NFS, NVMe, AWS S3

TECHNICAL PROJECTS & PROFESSIONAL DEVELOPMENT

- AI-Driven Web Application (OpenAI SDK Integration) | Personal Project / Continuing Education (2026)
 - Designed and deployed an interactive multi-turn agentic web application using the OpenAI Agents SDK, utilizing Gradio for the user interface.
 - Leveraged AI-assisted development ("vibe coding") and advanced prompt engineering to rapidly prototype, iterate, and optimize complex backend multi-turn agent loops.
 - Completed advanced, hands-on coursework via Udemy focused on production-level implementation of OpenAI SDKs, integrating these modern paradigms with core software architecture principles.
 - Example AI Web Application: https://huggingface.co/spaces/paulpowell/bible_xref
 - GitHub: https://github.com/ez-p/bible_xref